

● EASY

● ALGEBRA

● ANSWER KEY

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# Linear functions

# 01

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**Q1****C****C. 3**

Choice C is correct. If  $f(x) = \frac{2x-1}{3}$ , then  $f(5) = \frac{2(5)-1}{3} = \frac{10-1}{3} = \frac{9}{3} = 3$ .

Choice A is incorrect and may result from not multiplying  $x$  by 2 in the numerator. Choice B is incorrect and may result from dividing  $2x$  by 3 and then subtracting 1. Choice D is incorrect and may result from evaluating only the numerator  $2x - 1$ .

**Q2****9**

The correct answer is 9. It's given that  $gx = 6x$ . Substituting 54 for  $gx$  in the given function yields  $54 = 6x$ . Dividing both sides of this equation by 6 yields  $x = 9$ . Therefore, the value of  $x$  when  $gx = 54$  is 9.

**Q3****2**

The correct answer is 2. Substituting 8 for  $fx$  in the given equation yields  $8 = 4x$ . Dividing the left- and right-hand sides of this equation by 4 yields  $x = 2$ . Therefore, the value of  $x$  is 2 when  $fx = 8$ .

**Q4****D**

D. 740

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Choice D is correct. It's given that at the beginning of the 1st week of the year there was \$ 600 in a savings account and Gabriella deposits \$ 35 in that savings account at the end of each week. Therefore, the amount of money, in dollars, in the savings account at the end of the 4th week of that year is  $600 + 435$ , or 740.

Choice A is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the 4th week if Gabriella withdraws, rather than deposits, \$ 35 at the end of each week.

Choice B is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the 1st week, not the 4th week.

Choice C is incorrect and may result from conceptual or calculation errors.

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**Q5****C**

C. 700

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Choice C is correct. For the graph shown, the  $x$ -axis represents pressure, in atmospheres, and the  $y$ -axis represents temperature, in kelvins. Therefore, the estimated temperature, in kelvins, of the oxygen gas when its pressure is 6 atmospheres is represented by the  $y$ -coordinate of the point on the graph that has an  $x$ -coordinate of 6. The point on the graph with an  $x$ -coordinate of 6 has a  $y$ -coordinate of approximately 700. Therefore, the estimated temperature, in kelvins, of the oxygen gas when its pressure is 6 atmospheres is 700.

Choice A is incorrect. This is the pressure, in atmospheres, not the estimated temperature, in kelvins, of the oxygen gas when its pressure is 6 atmospheres.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

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**Q6**

**D**

D.  $y = x - 84$

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Choice D is correct. It's given that the number  $y$  is 84 less than the number  $x$ . A number that's 84 less than the number  $x$  is equivalent to 84 subtracted from the number  $x$ , or  $x - 84$ . Therefore, the equation  $y = x - 84$  represents the relationship between  $x$  and  $y$ .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

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**Q7****C**

C.  $f(x) = 4x + 1$

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Choice C is correct. Because  $f$  is a linear function of  $x$ , the equation  $f(x) = mx + b$ , where  $m$  and  $b$  are constants, can be used to define the relationship between  $x$  and  $f(x)$ . In this equation,  $m$  represents the increase in the value of  $f(x)$  for every increase in the value of  $x$  by 1. From the table, it can be determined that the value of  $f(x)$  increases by 8 for every increase in the value of  $x$  by 2. In other words, for the function  $f$  the value of  $m$  is  $\frac{8}{2}$ , or 4. The value of  $b$  can be found by substituting the values of  $x$  and  $f(x)$  from any row of the table and the value of  $m$  into the equation  $f(x) = mx + b$  and solving for  $b$ . For example, using  $x = 1$ ,  $f(x) = 5$ , and  $m = 4$  yields  $5 = 4(1) + b$ . Solving for  $b$  yields  $b = 1$ . Therefore, the equation defining the function  $f$  can be written in the form  $f(x) = 4x + 1$ .

Choices A, B, and D are incorrect. Any equation defining the linear function  $f$  must give values of  $f(x)$  for corresponding values of  $x$ , as shown in each row of the table. According to the table, if  $x = 3$ ,  $f(x) = 13$ . However, substituting  $x = 3$  into the equation given in choice A gives  $f(3) = 2(3) + 3$ , or  $f(3) = 9$ , not 13. Similarly, substituting  $x = 3$  into the equation given in choice B gives  $f(3) = 3(3) + 2$ , or  $f(3) = 11$ , not 13.

Lastly, substituting  $x = 3$  into the equation given in choice D gives  $f(3) = 5(3)$ , or  $f(3) = 15$ , not 13. Therefore, the equations in choices A, B, and D cannot define  $f$ .

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**Q8****B****B.** 40

Choice B is correct. For the graph shown, the  $x$ -axis represents time, in seconds, and the  $y$ -axis represents the number of candy bars wrapped. The slope of a line in the  $xy$ -plane is the change in  $y$  for each 1-unit increase in  $x$ . It follows that the slope of the graph shown represents the estimated number of candy bars the machine wraps with a label per second. The slope,  $m$ , of a line in the  $xy$ -plane can be found using any two points,  $x_1, y_1$  and  $x_2, y_2$ , on the line and the slope formula  $m = \frac{y_2 - y_1}{x_2 - x_1}$ . The graph shown passes through the points 0, 0 and 2, 80. Substituting these points for  $x_1, y_1$  and  $x_2, y_2$ , respectively, in the slope formula yields  $m = \frac{80 - 0}{2 - 0}$ , which is equivalent to  $m = \frac{80}{2}$ , or  $m = 40$ . Therefore, the estimated number of candy bars the machine wraps with a label per second is 40.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q9****D****D.** 902

Choice D is correct. Substituting 9 for  $x$  in the given equation yields  $f9 = 1009 + 2$ , or  $f9 = 902$ . Therefore, the value of  $f_x$  when  $x = 9$  is 902.

Choice A is incorrect. This is the value of  $f_x$  when  $x = 1.09$ .

Choice B is incorrect. This is the value of  $f_x$  when  $x = 1.16$ .

Choice C is incorrect. This is the value of  $f_x$  when  $x = 8.98$ .

## Q10

A

A. 0

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Choice A is correct. For the linear function  $f$ , it's given that  $f7 = 28$ . Substituting 7 for  $x$  and 28 for  $fx$  in the given function yields  $28 = 47 + b$ , or  $28 = 28 + b$ . Subtracting 28 from each side of this equation yields  $0 = b$ . Therefore, the value of  $b$  is 0.

Choice B is incorrect. Substituting 1 for  $b$  in the given function yields  $fx = 4x + 1$ . For this function, when the value of  $x$  is 7, the value of  $fx$  is 29, not 28.

Choice C is incorrect. Substituting 4 for  $b$  in the given function yields  $fx = 4x + 4$ . For this function, when the value of  $x$  is 7, the value of  $fx$  is 32, not 28.

Choice D is incorrect. Substituting 7 for  $b$  in the given function yields  $fx = 4x + 7$ . For this function, when the value of  $x$  is 7, the value of  $fx$  is 35, not 28.

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### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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